Date: | **Class:** | **Topic:**

xx-yy-zzzz | Science 101 | Some more examples for supported markdown structures [Letter format]

You can create/modify your own header styles too.

Top and bottom heights are limited (lower limits are 5mm):

- Top height limit: 80mm ~ 3.15″
  - usually used as header
- Bottom height limit: 100mm ~ 3.94″
  - usually used as summary

Chemistry Formulas:

$Fe^{II}Fe^{III}{}_2O_4$

$A \xrightarrow{H_2O} B$

$A{-}B{=}C{\equiv}D$ chem. bonding

$C_p[H_2O\,(l)] = 75.3\ \frac{J}{mol\,K}$

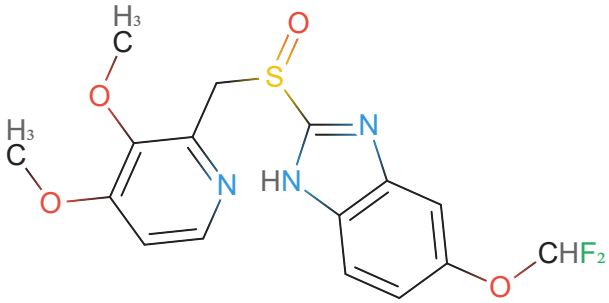
Math formula sizing by using standard $\LaTeX$ syntax:

1. `\tiny` – ca. **5–6 pt**
2. `\scriptsize` – ca. **7 pt**
3. `\small` – ca. **9 pt**
4. `\normalsize` – **10–12 pt** (not needed!)
5. `\large` – ca. **12–14 pt**
6. `\Large` – ca. **14–16 pt**
7. `\LARGE` – ca. **17–20 pt**
8. `\huge` – ca. **20–24 pt**
9. `\Huge` – ca. **25–30 pt** in MathJax.

Chemical Reaction Equations with mhchem Package: mhchem Syntax

$$Zn^{2+} \underset{+2\,H^+}{\overset{+2\,OH^-}{\rightleftharpoons}} \underset{\text{amphoteres Hydroxid}}{Zn(OH)_2 \downarrow} \underset{+2\,H^+}{\overset{+2\,OH^-}{\rightleftharpoons}} \underset{\text{Hydroxozikat}}{[Zn(OH)_4]^{2-}}$$

Molecules with SMILES:



The molecular formula of Pantoprazole is created with $\LaTeX$ code and the mhchem package: $C_{16}H_{15}F_2N_3O_4S$. The molecule pic was made using SMILES notation code. To get codes for both use a search engine with search terms: "Pantoprazole SMILES" and "Pantoprazole mhchem LaTeX". It's much faster and more accurate then to create it yourself. Copy & paste - it's done! For not so well known molecules ask your LLM.

A hint for beginners in bioinformatics, chemistry, biology and medicine:
Comparing the molecular formula count of atoms with the picture and SMILES code you might wonder why some atoms like hydrogen (H) or carbon atoms (C) are often missing? Here's why: SMILES simplifies molecular structures by omitting atoms. **In short, "missing atoms" in SMILES code from chemical databases and pictures are not errors – they are simplifications based on well-defined chemical rules and conventions**. These apply to other atoms too: nitrogen, oxygen, sulfur, halogens and even metals as well. All depends on conventions used in chemical databases or software interpreting the notation. Understanding these conventions is crucial for accurately interpreting SMILES strings. Getting code online from databases, Large Language Models (LLM's) or search engines is 99% reliable. So you really don't have to create the code by yourself on a regular base just to get your beautiful Cornell notes done.

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5. large-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

6. Large-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

7. LARGE-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

8. huge-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

9. Huge-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

1. Tiny:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (ad - bc)$

2. Script-size:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (ad - bc)$

3. Small-size:

$\sum_{i=1}^{n} i^2 = \frac{1}{2}n(n+1)$

4. Standard-size:

$\sum_{i=1}^{n} i^2 = \frac{1}{2}n(n+1)$

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua.

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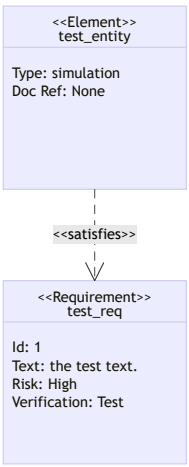
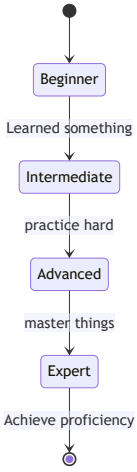
Tag-Pills for highlighting keywords:

#chemistry #biology #math #topics





My custom callout heading

- Lorem ipsum dolor sit
  - consetetur sadipscing elitr

- 1 custom type (choose Title and color with hex) syntax: **> [!custom|#1a20d9] My custom callout...**
- All 12 basic callout types (not the original Obsidian callout boxes, so with limited functionality)

Note
invidunt ut labore et dolore aliquyam erat

Abstract
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Info
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Failure
invidunt ut labore et dolore aliquyam erat

Tip
invidunt ut labore et dolore aliquyam erat

Danger
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Success
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Bug
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Question
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Example
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Warning
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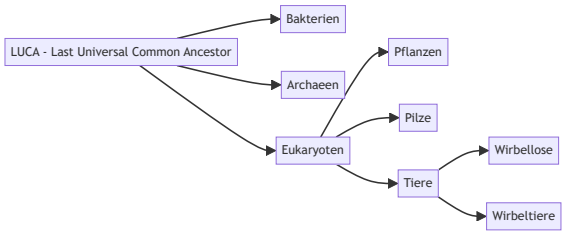
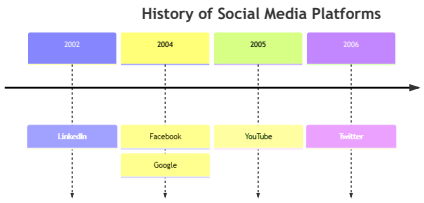
Quote
invidunt ut labore et dolore aliquyam erat

Some Mermaid Diagramms:

The syntax is quite easy to learn, but there are also graphical tools available and of course LLM's. Several software tools offer a **graphical user interface (GUI)** for creating Mermaid diagrams and exporting them as Mermaid code: **Mermaid Flow**, **Mermaid Chart**, **Miro** and **Eraser**